

EVOLUTION FROM ARTIFICIAL INTELLIGENCE TO LARGE LANGUAGE MODELS (LLMs)

EVOLUTION TIMELINE	1. EARLY AI (1950s – 1970s)	2. MACHINE LEARNING (1980s – 1990s)	3. DEEP LEARNING (2000s – 2010s)	4. TRANSFORMER ERA (2017 – 2020)	5. LARGE LANGUAGE MODELS (2020 – PRESENT)
KEY MILESTONES	1950: Turing Test 1956: Dartmouth Conference (coining of "AI") - Early rule-based systems (Logic Theorist, ELIZA)	1980s: Expert Systems 1986: Backpropagation 1990s: Statistical ML gains popularity (SVM, HMM, etc.)	2006: Deep Belief Networks 2012: AlexNet wins ImageNet 2014+: Deep Learning breakthrough in many domains	2017: "Attention Is All You Need" (Transformer) 2018: BERT 2019: GPT-2 2020: GPT-3	2020: GPT-3 (175B parameters) 2022: ChatGPT, PaLM, LLaMA 2023+: GPT-4, Claude, Gemini, Llama 3 and beyond
MAJOR ALGORITHMS / TECHNIQUES	- Rule-Based Reasoning - Search Algorithms (BFS, DFS, A*) - Knowledge Representation (Logic, Rules)	- Linear Regression - Decision Trees - Support Vector Machines - Naïve Bayes - k-NN, HMM, Clustering	- Artificial Neural Networks (ANN) - Convolutional Neural Networks (CNN) - Recurrent Neural Networks (RNN, LSTM, GRU)	- Transformer Architecture - Self-Attention Mechanism - Positional Encoding - Encoder-Decoder Models - Pre-training & Fine-tuning	- Large Transformers (Decoder-only & Encoder-Decoder) - Scaling Laws - Prompt Engineering - RLHF (Reinforcement Learning from Human Feedback)
ENABLING TECHNOLOGIES	- Basic Computing Power - Early Programming Languages (LISP) - Symbolic AI	- Increased Computational Power - Statistical Theory - Data Availability - Better Algorithms	- Big Data - GPUs & Parallel Computing - Advanced Optimization (Backprop, Adam) - Deep Neural Architectures	- Massive Datasets - High-Performance GPUs/TPUs - Transformer Innovation - Cloud Computing - Better Tokenization	- Extremely Large Datasets (Web, Books, Code, etc.) - Massive Compute (GPUs/TPUs) - Advanced Training Techniques - Human Feedback & Alignment
APPLICATIONS	- Theorem Proving - Game Playing (Early) - Expert Systems - Robotics (Rule-based)	- Spam Detection - Recommendation Systems - Credit Scoring - Speech Recognition (basic) - Computer Vision (basic)	- Image Recognition - Speech Recognition - Machine Translation - Autonomous Vehicles - Medical Diagnosis	- Advanced NLP Tasks - Text Summarization - Machine Translation - Code Generation (early) - Question Answering	- Chatbots & Virtual Assistants - Content Generation - Code Generation - Data Analysis & Insights - Education, Healthcare, Research - Automation & Decision Support

HOW THE TECHNOLOGIES BUILD ON EACH OTHER



★ **SUMMARY:** The journey from AI to LLMs is driven by advances in algorithms, data, and computing power. Each stage builds upon the previous one, leading to today's powerful Large Language Models that understand and generate human language at unprecedented scale.

E-Commerce AI System Concept Map

